

Antara Ganapathy

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Summary

Dedicated second-year BSc Computer Science student with a passion for tech-entrepreneurship, fin-tech and AI/ML. Having gained different skills and experience in the field, I aim to leverage my tech knowledge to drive social impact.

Education

The University of Manchester, BSc in Computer Science Sept 2024 – June 2027

- First-year courses: Mathematical Techniques for Computer Science, Fundamentals of Computer Engineering, Fundamentals of Computer Architecture, and Introduction to Programming I & II, Fundamentals of Computation, Data Science, Operating Systems
- Second-year courses (Semester 1): New Product Development and Innovation, Logic and modelling, Software Engineering, Introduction to AI, Programming Languages and Paradigms, Algorithms and Data Structures

Experience

Intern July 2025 - Sept 2025

Machani Robotics

- Building a Python-based interface enabling elderly users to play tablet games via Ria, a humanoid robot.
- Implementing Speech-to-Text and Text-to-Speech with AI for natural voice communication.
- Integrating Unity game with robot middleware to facilitate hands-free gameplay and user accessibility.

Spring into Software Engineering April 2025

JPMorganChase, Glasgow

- Attended panel discussions, networking sessions and workshops over the week to learn about Tech at JPMC
- Gained an in-depth understanding of the lines of businesses at JPMC and the role of tech in each
- Participated in the Code for Good Hackathon by developing PathMatch to improve the mentor registration and matching algorithm for MCR Pathways
- Developed PathMatch with React, Django, Tailwind & Gemini API

Projects

PathMatch Github

- Created PathMatch with my team as part of the Code for Good Hackathon held by JPMorganChase
- A streamlined service that aims to improve the mentor registration and matching algorithm for MCR Pathways.
- Developed with ReactJS, Tailwind, Django and Gemini API

Dish-it-Out Github

- Dish-it-Out is a smart grocery and kitchen management app made for people living in shared spaces.
- Includes features such as cost splitter, fridge manager, calendar system, chats between flatmates and an AI chatbot for recipe generation
- Developed with ReactJS and Tailwind CSS with the Toppy-M-7B model

Hack The Code Github

- Competition hosted by Reply
- Used Python to optimize the consumption of green energy resources
- Was in the top 20% scoring teams

RAG Chatbot Github

- Developed a chatbot with Retrieval-Augmented Generation to allow users to chat with a PDF

- Used Pinecone vector database and LangChain to develop the chatbot and Python Streamlit to develop the web-app

Sentiment Analysis

Github

- Trained a model with Natural Language Processing to analyze the sentiment news headlines
- Used web-scraping, with Python, to generate a test set of CNN news headlines

Glaucoma Detection

Github

- Trained a model to detect glaucoma on the 'Glaucoma Detection' Kaggle dataset
- Compared the performance of different pre-trained models: VGG, ResNet, Inception, and Xception
- 1/15 (of 200) students selected to present at the final round of the National High School Data Science Competition (NDSC), conducted in partnership with the Harvard Undergraduate Open Data Project

Fake News Detection

Github

- Trained a model using NLP to detect fake news using the 'Fake and real news dataset'.
- Implemented techniques such as the TF-IDF vectorizer and the multinomial naive Bayes classifier

Publications

Defining Evaluative Metrics for Medical Imaging Datasets

May 2023

Young, Scientist Journal Vanderbilt University

[Journal Link](#)

As society relies more on AI/ML in the medical field, it becomes imperative to get the dataset criteria right. This paper focuses on defining a metric to aid in the selection of a reliable medical imaging dataset.

Technologies

Python (Flask, FastAPI, AI/ML), Java, Dart (Flutter), HTML, CSS (Tailwind), JavaScript (React, Node), SQL, PHP